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Experiment - 1

Objective - Implement following commands used in Linux.

1. Ls :- It is used to list all files & directories in current directory.
For ex - 'ls -l' displays files & directories in current long dir listing format.

2. cal :- It is used to display calendar for specific month of year.
For Ex - 'cal 2022', shows full years calendar of 2022

3. tail :- It is used to display last few lines of a file.
For Ex - 'tail file.txt' displays last 10 lines of files.txt.

4. Man :- This command displays manual for other commands
For Ex :- 'man cal', displays manual for cal command.

5. Mkdir :- Used to create a new directory
For Ex :- 'mkdir manisha' creates a new directory 'manisha'.

6. more :- used to display contents of a file one page
For Ex - 'more file1.txt' displays content of 'file1.txt' one page at a time.

7. pwd :- Used to print the current working directory

8. rm :- It is used to remove files or directories
For Ex - 'rm file1.txt', remove file 'file1.txt' from the current directory.

9. less :- It provides more advanced features for navigating & searching through the contents of a file.

10.] who :- It is used to display who are currently logged into system

11] rmdir :- used to remove a directory

For Ex :- 'rmdir dir1', removes directory 'dir1'

12] cp :- used to copy files or directories.

For ex :- 'cp file.txt os.txt' called os.txt in current directory

13] whoami :- used to display the username of current used.

14] cat :- used to display content of file.

For ex :- "cat os.txt" display content of "os.txt" file.

15] mv :- used to move files or directories

for ex :- 'mv file.txt' dir1' move the file 'file.txt' to dir1 directory.

16] date :- It displays current date & Time

17] head :- It displays first few lines of file.

for ex :- 'head file.txt' displays first 10 lines of file.

18. echo :- used to display message in terminal window.

for ex :- 'echo "Hello"' displays "Hello" in terminal.

Perform following queries on basis of above commands.

[a] Who is current user?

```
who
```

[b] What is current login name?

```
whoami
```

[c] How to take backup of a file

```
cp file.txt file.txt.bak
```

[d] How to create 3 subdirectories in a directory using single line?

```
mkdir -p dir1/dir2/dir3
```

[e] How to remove file of .txt extension?

```
rm *.txt
```

[f] How to remove file of any extension.

Ans `rm *.*`

[g] How to remove file starting with 'a'.

```
rm a*.
```

[h] How to remove non-empty directory.

```
rm -r dir
```

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Experiment :- 2

Objective :- Implement following commands used in linux.

1] Cut :- Used to extract specific columns of text from a file or a stream of text.

For Ex - 'Cut -f2 file.txt > output.txt' Here it extracts the second column of 'data.txt' and store the result in another file 'output.txt'

2] last :- It shows a listing of last logged in users.

3] Uniq :- It filters adjacent matching lines in a file

For ex :- 'sort data.txt | uniq' Here it lists all unique lines of "data.txt".

4] Sort :- This command sort lines in a file in ascending or descending order.

For Ex - 'sort data.txt' it arrange data of file 'data.txt' in ascending order

5] Chmod - It is used to change the permission of files & directories.

For Ex - 'chmod 744 test.sh' This gives owner to read, write & execute the file & rest owner can only read & write

6] Sed :- Used to perform basic text transformation on an inspect file.

For Ex :- 'Sed' 's/unix/Linux/g' file.txt' Here all occurences of unix are replaced with Linux in 'file.txt'.

8] du :- shows the disk usage of files & directories.

9] grep :- It searches for a pattern in a file.

For Ex :- 'grep "friends" file.txt', Here it searches for the word "friends" in "file.txt".

9] diff :- It compares two files & shows the difference b/w them.

For Ex- 'diff file.txt file2.txt'

10] Pipe - The pipe (|) symbol is used to send output of one command as an input of other command.

For Ex :- 'Sort record.txt|uniq', it will sort the given file & print the unique vowels only.

11] History :- Displays a list of all previously executed commands.

Perform following queries on the basis of above commands.

a] Total no of users connected to system currently.

who / we -l

b] Display only current local time of system.

date + %T

c] Arranging the files of directory in ascending order of their size.

ls -ls

d] Display 5th character of your name.

echo "Krishna" | cut -c5

e] Display 7th line of a file.

sed -n '7p' file.txt

f] Sort a file on 2nd column basis

sort -k 2 file.txt

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Experiment-3

a] Write a shell script to display current date in a particular format, number of users currently logged in and current month's calendar.

Ans a

```
#current date in YYYY-MM-DD format
echo "Current date: $(date +%Y-%m-%d)"
#number of user currently logged in
echo "Number of users currently logged in: $(who | wc -l)"
#Current's month calendar
cal
```

b] Write a shell script to display the process and it's process id.

```
read -p "Enter the process name:" pname
pid = $(pgrep $pname)
echo "Process name: $ pname"
echo "Process ID: $pid"
```

c] Write a shell script to take name as input and display a greeting message to the user by checking.

```
read -p "Enter your name: " name
hour=$(date +%H)
#check the current time and display a greeting message
if [$hour -ge 5] && [$hour -lt 12]; then
  echo "Hello $name, Good Morning!"
elif [$hour -ge 12] && [$hour -lt 16]; then
  echo "Hell $name, Good Afternoon!"
else
  echo "Hello $name, Good Evening!"
fi
```

[d] Write a shell script to merge the content of 2 files into one file.

```
read -p "Enter the name of the first file: " file1
read -p "Enter the name of the second file: " file2
read -p "Enter the name of the output file: " file3
cat "$file1" "$file2" > "$file3"
echo "Merged contents of $file1 and $file2 into $file3"
```

[e] Write a shell script to create a tsv file containing name, roll no and age of 10 students. Then use that tsv file to display only the names of the students in alphabetical order.

```
echo -e "Name\t Roll No.\t Age" > students.tsv
echo -e "Manish\t 101\t 20" >> students.tsv
echo -e "Rishi\t 102\t 20" >> students.tsv
echo -e "Ayush\t 103\t 21" >> students.tsv
echo -e "Vaibhav\t 104\t 20" >> students.tsv
echo -e "Divyanshu\t 105\t 21" >> students.tsv
echo -e "Harshit\t 106\t 21" >> students.tsv
echo -e "Kunal\t 107\t 20" >> students.tsv
echo -e "Anuj\t 108\t 21" >> students.tsv
echo -e "Rahul\t 109\t 21" >> students.tsv
echo -e "Puneet\t 110\t 20" >> students.tsv
```

```
cut -f 1 students.tsv | tail -n +2 | sort
```

7. You are given a file of tab delimited weather data (TSV). There is no header column in this data file. The first five columns of this data are (a) the name of the city (b) The average monthly temperature in Jan (in Fahrenheit) (c) The average monthly temperature in April (in Fahrenheit) (d) The average monthly temperature in July (e) avg temp in Oct. You need to sort this file on the basis of average monthly temperature in April.

```
sort -t $'\t' -k 3 weather.tsv
```

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Experiment - 4

[a] Write a shell script that finds weather an entered number is even or odd.

```
echo "Enter the number"
read a
b= `expr $a % 2`
if [b -eq 0]
then
echo "$a is even"
else
echo "$a is odd"
fi
```

[b] Write a shell script to input the name of a file as command line argument and display wether it is a file, a directory or anything else.

```
filename=$1
if [-f "$filename"]; then
   echo "$filename is a file"
elif [-d "$filename"]; then
   echo "$ filename is a directory"
else
   echo "$filename is something else"
fi
```

---

save this file as file.sh
then in terminal
chmod 777 file.sh
./file.sh filename
#where filename is the actual filename or path to the file you want to check.

Q Write a shell script to input the marks of a student in 3 subjects and find his grade.

```
echo "Enter subject1 marks:"
read subj1
echo "Enter subject2 marks:"
read subj2
echo "Enter subject3 marks:"
read subj3
total=$((subj1+subj2+subj3))
percentage=$((total/3))
echo "Total marks are: $total"
echo "Percentage : $percentage%"
if [$percentage -ge 90]; then
   echo "Grade: A+".
elif [$percentage -ge 80] ; then
   echo "Grade A"
elif [$percentage -ge 70]; then
   echo "Grade B"
elif [$percentage -ge 60] ; then
  echo "Grade C"
  else
   echo "Grade: Fail"
  fi
```

[d] Write a shell script to accept a filename as argument and displays the last modification time if the file exists and a suitable message if it doesn't.

```
if [ -e "$1" ]; then
  echo "Last modification time of $1:"
  stat -c %y "$1"
else
  echo "$1 does not exist"
fi
```

save this file as file1.sh

Terminal

```
./file1.sh filename.txt
```

Replace 'filename.txt' with the actual filename or path to the file you want to check.

[e] Write the shell script to take file name as input and if the file exists then print the number of lines and also print 10th line of that line.

```
if [-e "$1"]; then
  echo "Number of lines in $1:"
  wc -l "$1"
  echo "10th line of $1:"
  sed -n 10p "$1"
else
  echo "$1 does not exist
```

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Assignment - 5

Q1 Find the day name when user input day number.

```
echo "Enter the number"
read a
case $a in
1)
echo "Sunday"
;;
2)
echo "Monday"
;;
3)
echo "Tuesday"
;;
4)
echo "Wednesday"
;;
5)
echo "Thursday"
;;
6)
echo "Friday"
;;
7)
echo "Saturday"
;;
*)
echo "Invalid Input"
;;
esac
```

Que 2 Check weather the input character is digit, vowel or character.

```
echo "Enter the input"
read a
case $a in
a|e|i|o|u|A|E|I|O|U|E)
echo "Entered number is vowel"
;;
[0-9])
echo "Entered number is digit
;;
*)
echo "other character"
;;
esac
```

Que 3 Check weather the number is even or odd

```
echo "Enter the number"
read a
b='expr $a%2'
case $b in
0)
echo "Entered number is even"
;;
*)
echo "Entered number is odd"
;;
esac
```

Que 4 Classify the number as single digit, 2 digit, 3 digit

```
echo "Enter the number"
read a
case $a in
[0-9])
 echo "Entered number is single digit"
 ;;
[0-9][0-9])
 echo "Entered number is two digit"
[0-9][0-9][0-9])
 echo "Entered number is three digit"
 ;;
esac
```

Assignment-6

Shell scripts related to loops and arrays.

[a] write a shell script to print multiplication table of a given no.

```
echo "Enter the number"
read n
a=1
while [$a -le 10]
do
b=$((n*a))
echo $b
a=$((a+1))
done
```

[b] write a shell script to implement a timer in Bash.

```
echo "Enter timer duration in seconds"
read duration
echo "Timer started for $duration seconds."
while [$duration -gt 0];
  do
  sleep1
  duration=$((duration-1))
  printf "\r%02d:%02d" $((duration/3600)).
                $(((duration/60)%60)) $((duration%60))

done
```

c] Write a shell script to print reverse of a given positive number.

```
echo "Enter the number"
read n
sum=0
while [ $n -gt 0 ]
do
sum=$((sum*10+n%10))
n=$((n/10))
done
echo "number in reverse is $sum".
```

d] write a shell script that print factorial of a given number.

```
echo "Enter the number"
read n
a=1
while [$n -gt 0]
do
a= $((a*n))
n=$((n-1))
done
echo "factorial is $a"
```

6] Write a shell script to demonstrate the working of array. Input 10 integer in array and display the maximum, minimum, total sum and average of an array

```
sum=0
echo "Enter 10 integers: "
for ((i=0;i<10;i++))
do
read arr[$i]
sum=$((arr[$i]+sum))
done
max=${arr[0]}
min=${arr[0]}
for i in "${arr[@]}"
do
if ((i>max))
then
    max=$i
fi
if ((i<min))
then
min=$i
fi
done
echo "Max value is:$max"
echo "Min value is:$min"
echo "Total sum is:$sum"
avg=$(echo "scale=2; $sum/10" |bc)
echo "Avg value is:$avg"
```

Output =>

```
./arr.sh
Enter 10 integers
1
2
3
4
5
6
7
8
9
10
Max value is:10
Min value is:1
Total sum is:55
Avg value is:5.50
```

F] You are given a file with four space separated columns containing the scores of students in three subjects. The first column contains a single character (A-Z) the student identifier. The next three columns have three number each the number are b/w 0 and 100 both inclusive these numbers denote the scores of the students in English, Mathematics and Science respectively. Your task is to identify whether each of the student has passed or failed. A student is considered to have passe if (s)he has score 50 or more in each of the three subject.

```
#read filename from command line argument
filename=$1
if [ ! -f $filename ]
then
    echo "file not found"
    exit 1
fi
#read in file line by line
while read line
do
   #split line into fields
   fields=($line)
  #Extract student identifier and score
   student = ${fields[0]}
   english = ${fields[1]}
   math = ${fields[2]}
   science = ${fields[3]}
```

```
#check if student has passed or failed
if [ $english -ge 50 ] && [ $maths -ge 50 ] &&
   [ $science -ge 50 ]
then
    result = "PASS"
else
    result = "FAIL"
fi

# print result for this student
echo "$student $result"
done < $filename.
```

Output:-

```
$ cat scores.txt
A 60 70 80
B 40 50 60
C 90 90 90
D 50 30 70

./arr.sh scores.txt
A PASS
B FAIL
C PASS
D FAIL
```